

Heather Leffew, PhD

AI Evaluation Scientist · Generative AI & LLM Systems · Behavioral Measurement

Applying the science of human behavior to the measurement of machine behavior

htmlleffew@gmail.com · DrHeatherLeffew.com · linkedin.com/in/heathertleffew

You should definitely check out the playable demo of a game I just built using a very cool, multi-level, hierarchical LLM engine with a built-in, live, AI-eval harness that tracks metrics as you play the game; [read about it](#) or [play it](#) - good luck, have fun!

EMPLOYMENT HISTORY

Director of Data Science

Sep 2025 – Present

Spokeo

Founding technical leader for Spokeo's Data Science organization, partnering with Product, Engineering, Legal, Compliance, and Executive Leadership to make production AI and ML systems measurable, auditable, and useful across identity, search, and conversational products for our ~20 million unique monthly users.

- Built and deployed evaluation frameworks for entity resolution, social identity, ranking, and conversational AI systems, including weak supervision, LLM-as-a-Judge, multi-model adjudication, and human-in-the-loop evaluation, grounded in construct validity and measurement theory.
- Designed and deployed an FCRA-compliant eval framework for SpokeoAI's conversational interface: replay-based evaluation validating ranking offline before live deployment, matching actual user next-action 87% of the time versus 14% for the prior system.
- Built an LLM-as-judge grading system for identity resolution where scalable ground-truth labels do not exist, encoding empirically measured platform characteristics into judge prompts, achieving 95.9% inter-rater agreement across human and LLM judges (8x cost reduction vs. human-only labeling).
- Architected a peer-reviewed deployment gating policy, catching a circular evaluation failure in label-provenance for a learning-to-rank model and a construct validity failure in an entity resolution model that had learned to group by data source rather than true indicators of shared identity.
- Introduced a Thin Root governance model separating probabilistic LLM runtimes from deterministic search policy, Replay-Based Evaluation loops quantifying conversational failure pathways without PII exposure, and a 4-tier FCRA compliance and ISO/IEC 42001 AI Management System.
- Designed custom agentic harnesses for AI-assisted workflows, embedding auditable review processes, evaluation evidence, and quality controls directly into engineering workflows.
- Led end-to-end delivery of a proprietary 15-step ML intelligence pipeline combining knowledge graphs, graph ML, and weak supervision for entity resolution and data fusion across ~49TB of multi-vendor data.

Full-Stack Principal Data Scientist & Applied Researcher

Jul 2022 – Jul 2025

TikTok USDS | ByteDance

Led applied ML research, evaluation strategy, and platform-integrity measurement across Trust & Safety, brand safety, election integrity, minor safety, and moderation systems for a platform with more than 180 million users in the US with annual revenue exceeding \$20 billion dollars. Owned the roadmap for ML measurement and detection capabilities, leading 25+ member cross-functional teams.

- Designed and deployed automated detection systems processing 1B+ records at 95% accuracy, enabling enforcement on 15K+ violative accounts with no detectable increase in appeal rates, a proxy for detection precision.

- Quantified over-enforcement as a measurable harm using NLP methods: discovered that >64% of posts flagged as harmful under a blunt legacy detection system were misclassified support and safety-promoting content, delivering improved methodology to balance precision and recall across all deployed detections.
- Built a comprehensive Content Moderation and Telemetry Suite applying Kaplan-Meier survival estimation, VADER sentiment, LDA topic modeling, NetworkX co-occurrence, and time-series anomaly detection, converting fuzzy safety questions into quantitatively defined behavioral and content typologies.
- Developed a harm-escalation detection and modeling system tracking violative content types and user discovery/consumption patterns across a 16-month retrospective study, identifying an early-intervention window for disrupting recommendation feeds while engaging platform policy development.
- Built election-integrity analytics unifying search, content, engagement, and moderation data into a shared evidence layer for cross-functional decision-making.
- Built distributed PySpark pipelines over nested moderation events to support user-level safety analysis, review prioritization, and forensic reporting with necessity, proportionality, accountability, and redaction guardrails.
- Led the creation of an industry-first external IRB methodology for ethical review and legal approval of behavioral research at platform scale.

Head of Accreditation

Nov 2020 – Jul 2022

National Emergency Responder and Public Safety Center

Partnered with the executive, sales, product, and content teams to lead accreditation, evaluation, and product operations for a public-safety training and certification organization. Managed the full accreditation lifecycle as we expanded nationally to serve more than 1,000 public safety agencies, leading to a multi-million dollar acquisition offer.

- Built standards-aligned assessment and review frameworks for emergency responder training, certification, renewal, and site-visit processes.
- Authored winning RFPs translating agency requirements into practical product and evaluation scope; presented risk analyses and accreditation findings to government officials and oversight bodies.

Director of Evaluations and Analytics

Jun 2019 – Jul 2022

Brower Psychological Police and Public Safety

Directed three cross-functional teams representing 15 individual contributors and 3 supervisors to design, deliver, and monitor digital psychological assessment products, people analytics platforms, and wellness programs in regulated public-safety environments.

- Built evaluation frameworks for psychological assessment, public-safety interventions, wellness programs, and organizational risk measurement.
- Increased operational throughput 5x through cloud-based digital transformation and grant-writing, securing an additional \$500K+ in grant funding, transforming the testing unit from a function operating at a loss, to the primary revenue driver for the organization, with increased client agency acquisition and reduced staffing needs.
- Developed predictive analytics for mental health service forecasting, applying evidence-based measurement to guide operational improvements.

Consulting Staff Data Scientist

Mar 2008 – Oct 2019

QMS Infotech, Inc.

Built analytics and audit products for private-sector hiring, workforce assessment, fairness review, and FCRA-compliant HR decision support. Partnered with B2B clients to identify systemic bias risks, strengthen assessment quality, and translate measurement findings into repeatable governance and remediation workflows.

- Designed analytics products evaluating hiring practices, workforce selection systems, adverse impact, and diversity outcomes across enterprise HR partners.
- Established psychometric and statistical standards for talent-assessment audits including construct validity, fairness measurement, and defensible decision criteria.
- Implemented bias-mitigating policies at scale that achieved full FCRA compliance for enterprise HR partners.

Practicum, Doctoral & Post-Doctoral Positions

May 2015 – Jul 2019

Fielding Graduate University

Conducted advanced psychological, diagnostic, neuropsychological, occupational, and forensic assessments across clinical, crisis, and public-safety populations. Supervised master's-level clinicians and practicum students.

- Delivered high-stakes clinical assessment, differential diagnosis, case conceptualization, psychosocial assessment, and forensic evaluation services across Cedar Springs Psychiatric Hospital, Rocky Mountain Behavioral Health, and Brower Psychological.
- Supported crisis intervention, fitness-for-duty evaluation, and disaster-response coordination where assessment quality carried operational consequences. Assistant Disaster Coordinator at Comprehensive Clinical Services (Aurora Mental Health Center).
- Contributed to research on clinically predictive linguistic analysis of Thematic Apperception Test narratives, supporting psychometric measurement of latent emotional and cognitive signals.
- Built the clinical foundation for later AI-evaluation work by applying validity, reliability, implicit measurement, and behavioral-risk reasoning in high-stakes human assessment settings.

Additional Professional Experience

Mar 2002 – Aug 2015

- **Neurotherapist**, *Unity Recovery Services, Inc.* – Managed neurotherapy delivery and complementary-treatment coordination for a substance-use recovery program. Delivered EEG neurofeedback, HRV biofeedback, audiovisual entrainment, and Alpha-Stim therapy. Designed treatment-modality suitability assessments and built intake-to-discharge symptom tracking evaluating changes in symptom type and severity across modalities.
- **Benefits Analyst**, *Florida Department of Children and Families* – Conducted eligibility determinations across state and federal assistance programs, balancing applicant interviews, policy interpretation, documentation review, and fraud identification affecting access to healthcare, food assistance, and emergency financial support. Prepared reports and data visualizations on program outcomes and service-delivery needs.
- **Consultant**, *Hope for Families* – Supported private-adoption operations through case management, regulatory compliance, stakeholder coordination, and documentation oversight. Strengthened practices for certification, accreditation, audit readiness, and regulatory compliance across legal, operational, and client-service aspects of the adoption process.
- **Professional Compliance**, *Broward Sheriff's Office* – Served in operational and compliance capacities combining emergency communications with audit, compliance, risk-management, and process-improvement across Court Services and Emergency Communications. Led digital transformation of audit systems improving structure, accessibility, and accreditation readiness. Conducted risk assessments and translated policy requirements into practical review workflows. Certified emergency communications cross-functional trainer.

EDUCATION

Doctor of Philosophy (PhD), Psychology – Quantitative Predictive Linguistics

Fielding Graduate University (APA Accredited)

Master of Arts (MA), Psychology – Advanced Psychometrics

Fielding Graduate University (WSCUC Accredited) | GPA 4.14

Bachelor of Science (BS), Psychology – Emergency Management & Public Safety

Liberty University (SACSCOC Accredited) | Summa Cum Laude | GPA 4.00

SELECTED PORTFOLIO PROJECTS

Dead Signal AI Evals – 7-evaluator harness attached to a playable generative narrative; per-turn automated scoring of coherence, persona fidelity, safety, and jailbreak resistance, running live in front of the reader.

Pathologizing Without Warrant – 26,158-observation NLP pipeline (HDBSCAN sense-disambiguation, three-layer voice segmentation, hand-coded validation) isolating 35 structurally uniform clinical role-violations in Claude Sonnet 4.5.

Reading Misalignment Off the Public Record – Discourse-mining pipeline (BERTopic over MiniLM embeddings, NetworkX co-occurrence, Prophet changepoint test) treating public user discussion as a standing misalignment instrument; demonstrated on Claude's harmful-helpfulness episode ($p < .001$).

A Recipe for Shipping AI Guardrails – Offline validation methodology for detection-and-control protocols: golden sets, adversarial sets, shadow replay, measured catch rate and over-refusal rate, all before live deployment.

Grading an Agent as a User Experience – Six-dimensional evaluation framework for tool-using agents: task success, grounding, tool-path efficiency, latency, language quality, and per-segment disaggregation.

The Constitution Your LLM App Already Has – Behavioral specification methodology: testable clauses, operator-vs-untrusted-input provenance, two-directional refusal measurement, and red-teaming the spec itself.

Instrumental and Affective Mass Murder: Establishing a Predictive Typology with Computer-Mediated Linguistic Analysis – 100-document corpus, LIWC2015 psychometric scoring, K-means clustering (100% cross-validated classification), logistic regression predicting attack characteristics from pre-attack language (McFadden $R^2 = 0.24$, OR = 14.11 on strongest predictor).

S K I L L S

AI Evaluation & Safety: AI Evals, LLM-as-Judge, Red-Teaming, Automated Red-Teaming Pipelines, Golden Datasets, Agentic Evals, Long-Horizon Reasoning Evaluation, Tool-Use Evaluation, RAG Evals, Simulation Testing, Hallucination Detection, Jailbreak Resistance Scoring, Prompt Injection Detection, Out-of-Character Behavior Detection, Context-Blindness Measurement, In-Character Consistency Evaluation, Hard-Fail Safety Gating, Evaluation Infrastructure & Frameworks, Scalable Oversight, Replay-Based Evaluation, Behavioral Taxonomies, Semantic Auditing, Epistemic Provenance, CUSUM Control Charts, Fixed Harness / Mutation Budget Architecture, Evaluation Gating, Adversarial Robustness, AI Alignment, Frontier AI Safety, Mechanistic Interpretability, Latent Evaluation Awareness, Activation Verbalization, Constitutional AI, 4×4 Refusal Matrix

Measurement & Research Methods: Psychometrics, Construct Validity, Criterion Validity, Convergent & Discriminant Validity, Test-Retest Reliability, Inter-Rater Reliability, Uncertainty Quantification, Experimental Design, A/B Testing, Causal Inference, Incrementality Testing, Lift Studies, Power Analysis, Source Independence Analysis, Ablation Studies, Behavioral Typologies, Item Response Theory, Classical Test Theory, Bayesian Uncertainty, Late Fusion Architecture, LIWC Methodology, Computational Linguistics, Psycholinguistics, HDBSCAN Clustering, BERTopic, Prophet Time-Series, NetworkX Co-occurrence

Machine Learning & Modeling: LLMs, Generative AI, Prompt Engineering, RLHF, Reward Modeling, Reinforcement Learning, CMDPs, Weak Supervision, Positive-Unlabeled Learning, Learning to Rank, Knowledge Graphs, Graph ML, Entity Resolution, Neuro-Symbolic AI, XGBoost, Logistic Regression, Survival Analysis, Semantic Clustering, NLP, Topic Modeling, Sentiment Analysis, Synthetic Data Generation, SHAP Explainability, Isotonic Calibration, MediaPipe, Whisper

MLOps, Platform & Infrastructure: Model Context Protocol (MCP), End-to-End ML Pipeline Architecture, Model Lifecycle Management, Distributed Computing, Apache Spark, PySpark, Databricks, Delta Lake, AWS SageMaker, AWS Glue, AWS EMR, Airflow, Docker, Kubernetes, CI/CD, DuckDB, PyArrow

Governance, Safety & Compliance: Responsible AI, AI Governance, ISO/IEC 42001, NIST AI RMF, EU AI Act, FCRA Compliance, Bias Mitigation, Algorithmic Auditing, Privacy-Preserving ML, RBAC, Trust and Safety, Content Moderation, Platform Integrity, Over-Enforcement Measurement

Technical Stack: Python, PySpark, SQL, Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers, Scala, Kafka, Hive, Tableau, Power BI, Databricks MLflow, AWS ML Services, HTML, CSS, JavaScript, Astro SSG, Firebase